



POLITÉCNICA

Information Security
Block II
Elliptic Curves and Hash Functions



Goals

Getting ready for the Block II exam:

- Elliptic Curves
- Hash Functions

Elliptic Curves

Exercise 1. Write the formula for $E_{17}(1,3)$ and check which of the following points belong to the curve.

(8,8)

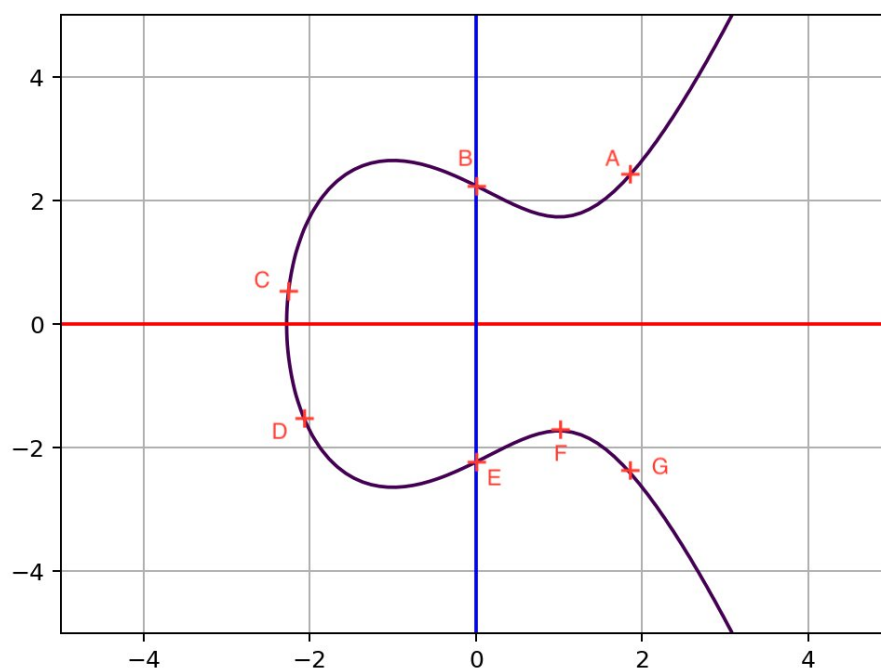
(2,9)

(6,6)

(11,6)

Exercise 2. Given the following elliptic curve, calculate, graphically, the following operations:

- $A+C$
- $2F$
- $D+E+F$
- $B+E$
- $G+D$
- $4A$
- $3F$



Exercise 3. Working on the curve $E_{11}(1,0)$ complete a Diffie-Hellman key exchange and obtain the agreed key. Use the following parameters for the exchange:

Generator $G = (7,8)$

Alice's secret value = 2

Bob's secret value = 3

Useful data about the curve:

11 points + O . $\{O, (0,0), (5,3), (5,8), (7,3), (7,8), (8,5), (8,6), (9,1), (9,10), (10,3), (10,8)\}$

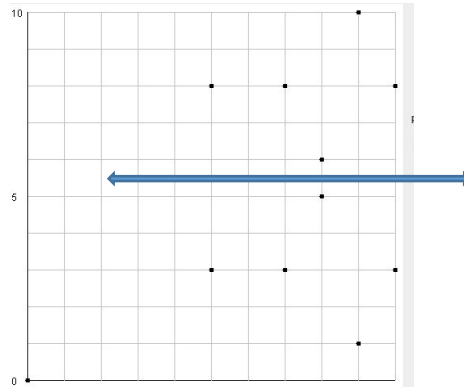


Table of sums

$+$	O	(0,0)	(5,3)	(5,8)	(7,3)	(7,8)	(8,5)	(8,6)	(9,1)	(9,10)	(10,3)	(10,8)
O	O	(0,0)	(5,3)	(5,8)	(7,3)	(7,8)	(8,5)	(8,6)	(9,1)	(9,10)	(10,3)	(10,8)
(0,0)		O	(9,10)	(9,1)	(8,6)	(8,5)	(7,8)	(7,3)	(5,8)	(5,3)	(10,8)	(10,3)
(5,3)				O	(10,8)	(8,6)	(7,3)	(10,3)	(0,0)	(9,1)	(7,8)	(8,5)
(5,8)				(5,3)	(8,5)	(10,3)	(10,8)	(7,8)	(9,10)	(0,0)	(8,6)	(7,3)
(7,3)					(9,10)	O	(0,0)	(5,3)	(7,8)	(10,3)	(5,8)	(9,1)
(7,8)						(9,1)	(10,3)	(0,0)	(10,8)	(7,3)	(9,10)	(5,3)
(8,5)							(9,1)	O	(10,3)	(8,6)	(5,3)	(9,10)
(8,6)								(9,10)	(8,5)	(10,8)	(9,1)	(5,8)
(9,1)									(5,3)	O	(7,3)	(8,6)
(9,10)										(5,8)	(8,5)	(7,8)
(10,3)											(0,0)	O
(10,8)												(0,0)

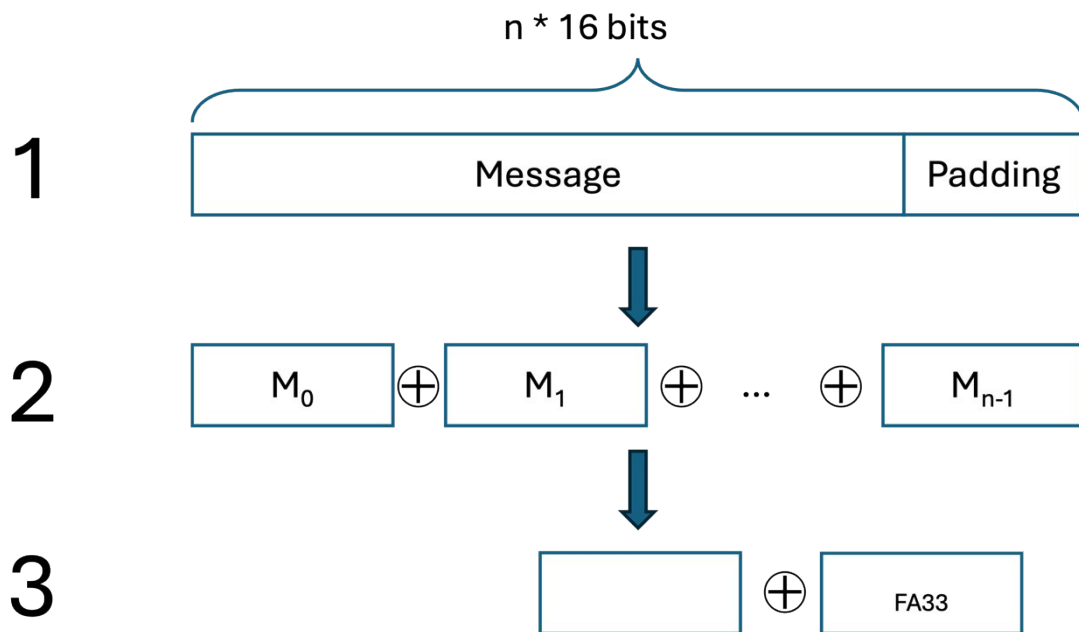
Hash Functions

Exercise 1. Describe the minimum number of tries (on average) that would be required when executing a birthday paradox attack against the following hash functions:

- MD5
- SHA1
- SHA2 con 512 bits
- SHA2 con 256 bits

Exercise 2. Given a 16 bit Hash function that works in the following way.

1. Add padding until the message length is multiple of 16 bits
2. Split the message in 16-bit blocks and perform the XOR operation between each 16-bit block consecutively. The result should be a 16-bit block
3. Perform the XOR operation between the resulting block and the constant (FA33 in hexadecimal - 1111 1010 0011 0011 in binary).



Answer the following questions, justifying all your answers:

- Does this hash function follow the diffusion property?
- Does this function follow the compression property?
- Is it resistant to preimage attacks?
- Find two messages that result in the same hash